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CENTRAL INSTRUMENTATION CENTER



SPRAY DRYER



In the world of industrial dryers, there are few types that accept pumpable fluids as the feed material at the inlet end of the process and produce dry particulate at the outlet. Spray drying is unique in its ability to produce powders with a specific particle size and moisture content without regard for the capacity of the dryer and the heat sensitivity of the product. This flexibility makes spray drying the process of choice for many industrial drying operations.

Rotary atomizers use the energy of a high speed rotating wheel to divide bulk liquid in to droplets. Feedstock isintroduced at the center of the wheel, flows over thesurface to the periphery and disintegrates into dropletswhen it leaves the wheel.

ADVANTAGES OF ROTARY ATOMIZERS:

- Because the break-up energy is supplied by the motordriven wheel, the liquid feed system can operate atrelatively low pressure.
- Able to handle applications where clogging would bea problem for nozzles.
- Particle size can be changed by changing the wheelspeed.

DIFFERENTIAL SCANNING CALORIMETER (DSC)



Calorimetry is a primary technique for measuring the thermal properties of materials to establish a connection between temperature and specific physical properties of substances and is the only method for direct determination of the enthalpy associated with the process of interest. Calorimeters are used frequently in chemistry, biochemistry, cell biology,6 biotechnology, pharmacology, and recently, in nanoscience9 to measure thermodynamic properties of the bio molecules and nano-sized materials. Amongst various types of calorimeters, differential scanning.

DSC is a thermal analysis apparatus measuring how physical properties of a sample change, along with temperature against time. In other words, the device is a thermal analysis instrument that determines the temperature and heat flow associated with material transitions as a function of time and temperature. During a change in temperature, DSC measures a heat quantity, which is radiated or absorbed excessively by the sample on the basis of a temperature difference between the sample and the reference material.

FREEZE DRYER (LYOPHILIZER)

Lyophilizer and freeze dryer are synonymous names for the same equipment. A lyophilizer executes a water removal process typically used to preserve perishable materials, to extend shelf life or make the material more convenient for transport. Lyophilizers work by freezing the material, then reducing the pressure and adding heat to allow the frozen water in the material to sublimate.

A Lyophilizer's 3 Primary Stages

A lyophilizer functions in three phases, with the first and most critical being the freezing phase. Proper lyophilization, otherwise known as freeze drying, can reduce drying times by 30%.

Freezing Phase

A lyophilizer uses various methods to freeze the product. Freezing can be done in a freezer, a chilled bath (shell freezer), or on a shelf in the lyophilizer. The lyophilizer cools the material below its triple point to ensure that sublimation, rather than melting, will occur. This preserves the material's physical form.

A lyophilizer most easily freeze dries large ice crystals, which can be produced by slow freezing or annealing. However, with biological materials, when crystals are too large they may break the cell walls, and that leads to less-than-ideal freeze drying results. To prevent this, the freezing is done rapidly. For materials that tend to precipitate, annealing can be used. This process involves fast freezing, then raising the product temperature to allow the crystals to grow.

Primary Drying (Sublimation) Phase

A lyophilizer's second phase is primary drying (sublimation), in which the pressure is lowered and heat is added to the material in order for the water to sublimate. The lyophilizer's vacuum speeds sublimation. The lyophilizer's cold condenser provides a surface for the water vapor to adhere and solidify. The condenser also protects the vacuum pump from the water vapor. About 95% of the water in the material is removed in this phase. Primary drying can be a slow process. Too much heat can alter the structure of the material.



Secondary Drying (Adsorption) Phase

A lyophilizer's final phase is secondary drying (adsorption), during which the ionically-bound water molecules are removed. By raising the temperature higher than in the primary drying phase, the bonds are broken between the material and the water molecules. Freeze dried materials retain a porous structure. After the lyophilizer completes its process, the vacuum can be broken with an inert gas before the material is sealed. Most materials can be dried to 1-5% residual moisture.

PARTICLE SIZE ANALYZER AND ZETA POTENTIAL

The DLS Particle Size and Zeta Potential Analyzer is a user-friendly system for colloidal, nano particulate and macro moleculer characterization. It can determine particle size distribution, particle zeta potential (related to the magnitude of the electrical charge at the particle surface) and molecular weight of large polymeric substances dispersed in water. The significance of zeta potential is that its value can be related to the stability of colloidal dispersions (e.g., a multivitamin syrup). The zeta potential indicates the degree of repulsion between adjacent, similarly charged particles (e.g., the vitamins) in a dispersion. For molecules and particles that are small enough, a high zeta potential will confer stability, i.e., the solution or dispersion will resist aggregation. When the potential is low, attraction exceeds repulsion and the dispersion will break and flocculate. Dynamic Light Scattering (DLS) principle can be used to determine the size distribution profile of small particles in suspension or polymers in solution. It can also be used to probe the behaviour of complex fluids such as concentrated polymer solutions.



- Particle Size and Molecular Size: It can measure size between 0.3 nm to 6 µm with a minimum volume 10 µL of sample suspension. It claims an accuracy of ±2% with a precision of ±1%.
- Zeta Potential: It can measure wide range of zeta potentials for particles in the size range 1 nm – 100 µm, depending on the type of sample. Minimum sample volume required is 175 µL. It claims an accuracy of 0.12 µm.cm/V.s and a sensitivity of 10 mg/mL (BSA). It can analyse a wide range of suspension concentrations (0.1 ppm to 40% v/v depending on the sample type).
- Molecular Weight of polymers dispersed in solution: On a sample volume of 10-40 µL, the instrument can determine a range between 1kDa to 25 MDa. Accuracy < ±10%
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MAX-XP ULTRACENTRIFUGE

The Optima MAX-XP microprocessor-controlled tabletop ultracentrifuge (see Figure) generateshigh centrifugal forces for a variety of applications. The ultracentrifuge design features a variablefrequency induction drive, thermoelectric temperature control system, self-purging vacuum system, rotor overspeed identification system, user login feature, program memory that contains multiple five-step programs, and a choice of acceleration and deceleration rates. Manual and programmed operations are available from the integrated touchscreen interface.



- In manual operation, you enter the individual run parameters before beginning each run.
- In programmed operation, you can duplicate runs quickly and accurately by selecting previously entered programs and running them again. Additionally, a pulse feature allows you to manually accelerate the rotor for sample preparation.

This ultracentrifuge is designed to be installed by a Beckman Coulter Field Service representative. Installation by anyone other than authorized Beckman Coulter personnel invalidates any warranty covering the ultracentrifuge.

POWDER FLOW TESTER

Teledyne LABS Flodex™ tester provides a simple technique for assessing a broad range of materials in powder form.
The Flodex is an easy-to-use tool for determining the flow ability index of a powder over an arbitrary scale of 4 to 40. This index can be used as a specification for raw materials. The test takes into account numerous parameters that affect powder flow ability, such as:

Particle size and shape
"Fines"
Unit surface area
Actual density and bulk density
Porosity
Settling characteristics
Electrostatic charges



The Flodex intrinsic flow ability method is based on the ability of a powder to fall freely through an orifice in a disk. The system comes with a set of 19 disks with orifice sizes ranging from 4 mm to 34 mm. When used in pharmaceutical manufacturing applications the Flodex system is fully compliant with USP <1174> Powder Flow.

ROTARY EVAPORATOR WITH VACUUM AND CHILLER

Rotary evaporators work on the principle that solvents have a range of boiling points, which decrease under reduced pressure. The evaporation flask rotates at a specified speed forcing the materials to form a large area of thin film on the inner wall of the flask.



Key Benefits of Using a Rotary Evaporator

Users can carry out solvent removal faster and more efficiently than evaporation under atmospheric pressure, thus saving time and increasing productivity in the laboratory. In part, this accelerated evaporation is due to the film's formation on the flask's inner surface, which increases evaporation significantly. It is therefor relatively quick to achieve solvent removal (depending on volume and solvent). Rotary evaporators are simple to use and are commonly found in laboratories.

The flask is heated evenly, and materials with a lower boiling point rapidly evaporate. Recycling of the solvent stream occurs in the receiving flask, following cooling by the glass condenser.

DIGITAL PLETHSYMOGRAPH

The primary purpose of this monograph is to present the technical details associated with a method of recording the rate of blood flow into and out of the digits of man. The author describes the various theoretic considerations involved in obtaining accurate readings, using the venous occlusion plethysmographic method. Attention is also called to the dangers of error encountered and to the means used to minimize or evade them. Through an analysis of the curves obtained from the finger with a double-beam plethysmograph, the author is able to measure the time course of the rate of blood into and out of the finger tip.

The Digital Water Plethysmometer is designed to provide a highly useful tool in the measurement of small volume changes. This test is typically used to follow the evolution of the inflammatory response experimentally induced in rodents and to screen potential anti-inflammatory or anti-edema properties of pharmacological substances.



Basically, the volume transducer is formed by two Perspex tubes interconnected and filled with a conductive solution and a platinum electrode for each chamber. All the system is supported by a stand (included) that can be placed over the control unit.

The water displacement produced by the immersion of the animal paw in the measuring tube is reflected into the second tube, inducing a change in the conductance between the two platinum electrodes. The Plethysmometer Control Unit detects the conductance changes and generates an output signal to the digital display indicating the volume displacement measured (0.01 ml resolution). The current value remains in the digital display until a new trial starts. The Control Unit is automatically zeroed between successive readings, thus making intermediate adjustments unnecessary.
A remote foot-switch allowing rapid hands-free experiments can be used to set control the end point of the measurement.

AUTO ANALYZER

An automated analyser is a medical laboratory instrument designed to measure various substances and other characteristics in a number of biological samples quickly, with minimal human assistance. These measured properties of blood and other fluids may be useful in the diagnosis of disease.

Photometry is the most common method for testing the amount of a specific analyte in a sample. In this technique, the sample undergoes a reaction to produce a color change. Then, a photometer measures the absorbance of the sample to indirectly measure the concentration of analyte present in the sample. The use of an ion-selective electrode (ISE) is another common analytical method that specifically measures ion concentrations. This typically measures the concentrations of sodium, calcium or potassium present in the sample.



There are various methods of introducing samples into the analyser. Test tubes of samples are often loaded into racks. These racks can be inserted directly into some analysers or, in larger labs, moved along an automated track. More manual methods include inserting tubes directly into circular carousels that rotate to make the sample available. Some analysers require samples to be transferred to sample cups. However, the need to protect the health and safety of laboratory staff has prompted many manufacturers to develop analysers that feature closed tube sampling, preventing workers from direct exposure to samples. Samples can be processed singly, in batches, or continuously.
The automation of laboratory testing does not remove the need for human expertise (results must still be evaluated bymedical technologists and other qualified clinical laboratory professionals), but it does ease concerns about error reduction, staffing concerns, and safety.

RT PCR EQUIPMENT WITH ACCESSORIES

Real-time PCR thermal cyclers, or qPCR machines, are analytical instruments that amplify DNA and measure the resulting fluorescence generated by amplification. Real-time PCR instruments measure signals generated by fluorescent probes that are proportional to DNA amplification, allowing accurate quantification of specific sequences. These devices are crucial for enabling experiments in applications such as gene expression, genetic variation, genotyping, biomarker discovery, and specific detection of rare targets, bacteria, and viruses. These specialized instruments are capable of quantifying very small amounts of DNA with good dynamic range.



In addition, data can be readily analyzed without post-PCR processing, such as gel electrophoresis. In selecting a qPCR machine, keep an eye out for helpful features, including individually programmed wells, increased capacities for fluorescent color detection, faster run times, easily interchangeable blocks, automation options, remote connectivity, and access to cloud computing. Various configurations are offered, such as 2 x 48 wells, 96 wells, or 384 wells, which serve to accommodate the unique needs of laboratory groups. Browse the latest in qPCR

PARALLEL SYNTHESIZER

Parallel synthesis enables substantial time savings and compound differentiation by running multiple experiments simultaneously.

Parallel synthesis is used to accelerate the discovery of new compounds and to screen for optimal process conditions. In the pharmaceutical industry, parallel synthesis is used for the discovery and development of potential drug candidates. For example, this enables the synthesis of libraries with diverse chemical structures that can be screened for potential biological activity.

Parallel Synthesis applications run at different scales, including lead generation, lead optimization, and screening for optimal reaction conditions. Thus, in process scale-up and development, parallel synthesis methods speed process optimization by providing a better understanding of the effect of reaction variables including solvent systems, optimal temperatures and concentrations, correct reagents, reaction times, and selection of catalysts..



In drug discovery, a target molecule, such as a protein, is identified which has been associated with a disease process. Once the target molecule has been validated, scientists initiate a discovery process to find other molecules or substances that act on the target. The process of finding a molecule (i.e. lead molecule) that acts on a target can be daunting; thousands of compounds are likely needed to be screened to find one or more promising leads. Once a lead molecule is identified, it can be chemically modified to improve its efficacy against the target molecule.

Parallel synthesis is vital in the step of screening for lead compounds. Parallel synthesis enables the rapid, logical synthesis of thousands of molecules, any one of which could provide a hit against the target entity. Prior to the use of parallel synthesis in combinatorial chemistry, the discovery of a lead molecules was a sequential process that was time consuming and/or fortuitous.

CATALYST MICROWAVE

Microwave radiationis a proven technology to enhance the efficiency of organic synthesis. The selective absorption of radiation by polar compounds and conversion of electromagnetic energy into heat bydielectriclosses forms the basis of the nonconventional microwavedielectric heating. This technology facilitates the acceleration of the reaction rate and enhancement in the product yields through uniform and selective heating thereby; it aids in the development of cleaner and greener synthetic routes. The initial part of the chapter is more comprehensive and discusses the theoretical background, microwave mechanisms, and role of microwaves in differentorganic reactions. The latter part of the chapter discusses briefly the types of microwave reactors, current challenges in microwave-assisted synthesis regarding energy efficiency, and scale-up of organic transformations.



Microwave-assisted synthesis has attracted wide attention as an efficient and energy-saving synthesis strategy due to its unique heating mechanism. Environmental catalysts prepared via microwave heating process typically exhibit uniform crystal size with desired morphology, porous structure and interfacial properties, which are important for the catalytic remediation of air pollution. This review first summarizes the principles of microwave heating, followed by the characteristics of the obtained catalysts. The research progress of representative materials (e.g., supported metal catalysts, metal oxides, and porous carbon materials) in the treatment of gaseous pollutants are then categorized in detail. Meanwhile, this review compiles the recent research outputs of air pollutants remediation by environmental catalysts from the perspective of features and reaction pathways facilitated by microwave-assisted synthesis.

It demonstrates that microwave-assisted heating can not only improve the synthesis efficiency and the catalytic activity but also modify the selectivity by tuning the catalyst properties. Finally, the use of microwave irradiation to activate environmental catalytic reactions is reviewed, as the application of microwaves can lead to significant improvements due to the possibility of delivering energy directly to the catalytic materials. Nevertheless, challenges remain in the development of highly active and efficient catalyst to realize its practical application. These, together with prospects on the development of environmental catalysts using microwave techniques are provided for enlightening the future of this field of research.

GAS SENSING UNIT

Biologists think the human nose can distinguish about 10,000 odors. A dog's nose can distinguish about 2 million different odors, and it also has a high "analytical ability". A specific smell can be found from many mixed smells. Even so, there are still many colorless and odorless gases in nature that neither humans nor animals can smell. Therefore, we use gas sensors to measure the concentration of different gases.

A gas sensor detects a specific gas and measures whether the gas is present near the sensor, or its content in the air. Therefore, gas sensors are often indispensable in safety systems. The gas sensor has many different types and can measure different gases. There are sensors that measure non-toxic gases such as oxygen and carbon dioxide, and sensors that detect toxic gases such as carbon monoxide, TVOC, and ammonia. Gas sensors are often used in flammable, explosive and toxic places to protect workers.



There are many gas sensors types, in order to choose the most suitable gas sensor, we need to understand the various sensor characteristics. For example: Which sensor can measure toxic gases? Which sensor is convenient to carry? Which sensor measures more accurately? If you know enough about the sensor, you can definitely make the right choice.

According to different gas types: it can be divided into combustible gas sensors (often using catalytic combustion, infrared, thermal conductivity, semiconductor type), toxic gas sensors (generally using electrochemical, metal-semiconductor, photoionization, flame ionization), Harmful gas sensors (often infrared, ultraviolet, etc.), oxygen (often paramagnetic, zirconia) and other types.

According to different usage methods: it can be divided into portable gas detectors and fixed gas detectors.

According to different sampling methods: it can be divided into diffusion sensors (the sensor is directly installed in the measured environment, and the measurement gas is in direct contact with the detection unit through natural diffusion), pumping sensors (through the suction pump, etc., the gas is sucked into the detection unit. According to whether the gas needs to be diluted, it is divided into complete inhalation and diluted inhalation).

According to different functions: it can be divided into a single gas sensor (only one gas can be detected) and a composite gas sensor (can detect multiple gases at the same time).

According to the different detection principles: it can be divided into semiconductor gas sensor, electrochemical gas sensor, NDIR gas sensor, catalytic gas sensor, thermal conductivity gas sensor, magnetic gas sensor and so on.

ULTRASONIC PROBE

Sonicators are high-frequency tools (20 kHz) that use ultrasonic energy to agitate particles in liquids. These devices are employed to facilitate a wide variety of processes, such as mixing, cleaning, degassing, cell disruption, and sample preparation.

Note that when sound travels through air, solids, and liquids in the form of sound waves, sound energy is manifested as vibration, with sound waves with higher frequency causing more vibrations. Vibrations in solutions, on the other hand, lead to cavitation – or the formation of vacuum bubbles – which allows sonicators to facilitate dissolution, homogenization, milling, cell lysis, chemical reactions, degassing, deagglomeration, and cleaning.

Sonication devices are indispensable in research and production settings. Note that such devices can be divided into bath sonicators and probe sonicators. Bath sonicators are ultrasonic tools that spread energy diffusely, making them invaluable for the processing of large volumes, cleaning, sterilizing, and degassing. As bath sonicators do not require a lot of power, they are more durable and affordable units. That said, bath sonicators can't be employed for small particle sizes, and due to the uneven energy they produce, they do not support replication.



Probe sonicators, on the other hand, allow the particles around the probe to be blasted with high levels of energy, which is ideal for size reduction, cell disruption, emulsification, and dispersion of nanoparticles. Note that these devices are also called cell disruptors and ultrasonic homogenizers. Even though probe sonicators are highly sophisticated units (usually more costly than bath sonicators), we should note that erosion of the probe tip might lead to cross-contamination.

Sonicators are vital lab tools used for a wide range of applications, such as degassing, mixing, cleaning, cell disruption, and dissolution. By using sound energy, sonicators allow no-touch manipulations of different samples. Before buying a sonicator, users should consider three major factors: requirements, specifications, and total costs. Here we should note there are two types of sonicators: bath sonicators that are ideal for cleaning and degassing, and probe sonicators suitable for cell disruption and nanoparticle dispersion, with probe sonicators being more costly than ultrasonic baths. When it comes to probe sonicators, users should consider purchasing different accessories, such as booster horns and different probe tips for different volumes and sample types. Maintenance and safety are also crucial to ensure a sonicator's long life and accurate use.

CHNS(O) ANALYZER

A CHNSO analyzer, also known as an elemental analyzer, gives a quantitative measure of elemental composition, that is, organic elements of a given sample, and also allows the deciphering of the empirical molecular formula of organic substances. The CHNSO Analyzer has multiple uses in many industries and sectors, for example the food and pharmaceutical industry.

A CHNSO analyzer is a sophisticated instrument designed for the precise determination of the elemental composition of organic compounds. It consists of several key components, each playing a crucial role in a CHNSO analyzer:



CHNSO Analyzer Sample Chamber and Sample Preparation

The sample chamber of a CHNSO analyzer is where the organic sample to be analysed is introduced into the instrument. Samples need to be prepared carefully to ensure accurate results.

CHNSO Analyzer Combustion Unit:

The CHNSO analyzer combustion unit is a central component of the CHNSO analyzer, responsible for breaking down the organic sample into its constituent elements through combustion analysis. It operates by heating the sample in the presence of oxygen (O2), leading to the conversion of carbon (C) and hydrogen (H) into carbon dioxide (CO2) and water (H2O), respectively.

Nitrogen (N) is converted to nitrogen oxides (NOx and then into nitrogen gas (N2), and sulphur (S) is transformed into sulphur dioxide (SO2). The CHNSO analyzer combustion unit's significance lies in its ability to convert the organic material into forms that can be quantified accurately, allowing for the determination of C, H, N, S, and O content.

VICKER'S MICRO-HARDNESS TESTER

Micro Vickers Hardness Testers are capable of accurate measurement of hardness numbers using state-of-the-art image processing technology. They can test a variety of products from soft metals to hardened steel with high accuracy. For loading and unloading, they have a motorized cycle and for load selection, manual machines have external knobs.

FEATURES



- Horizontally rotating turret with four slots for magnification lenses and two for indenters
- Indent Reading
- Automatic focus with auto reading
- Automatic focus with auto reading
- Manual focus with auto reading
- Load forces are applied through load cells and controlled in "Closed Loop" with a frequency of 1 kHz, assuring perfect linearity in every range
- Motorized table 100 x 60 mm or 200 x 100 mm (automatic) with 0.5 µm step
- Test Loads: from 1 gf to 2 kgf (from 0,0098 to 19,614 N)
- The ergonomic and handily wheel on the tester side controls the Z stroke
- The vertically sliding chromed spindle provides stable support for the specimen
- Perform reliable tests even on multilevel or misaligned sample surfaces

WIRE CUT ELECTRO-DISCHARGE MACHINE

Wire Cut EDM provide a wide range of opportunities and possibilities in cutting intricate shape and size with precise cutting speed, giving a clear advantage over conventional cutting equipment's. In this process, a thin Mo metallic wire of diameter 0.18 mm is fed on to the workpiece which is held in a tank and the deionized water is used as a coolant fluid. The wire is fed continuously through the spool between the upper and lower guide ways. The sample which is cut will have high heat and temperature which melts and remove the workpiece for required dimensions.

MICRO PLASMA ARC WELDING MACHINE

In Micro Plasma Welding, a second nozzle is used in addition to shielding gas nozzle. Through this second nozzle, plasma flows, surrounding and constricting the arc, which leads to a high power density and stability. The surrounding gas protects the melt bath of oxidation.

ADVANTAGES OF PLASMA ARC WELDING



- Higher welding speed and less distortion as well as a higher insensitivity regarding changes in distance.
- Due to the stabilizing effect of the plasma it is possible to work even with low currents. This makes micro plasma welding suitable for joining very thin metal sheets.
- Less distortion.
- Models capable to weld sheets from 100 microns to 1 mm thickness.
- Built-in Pilot Arc system.
- Pulsing facility available for perfect and smooth welding.

ROBOTIC ARM MIG WELDING SYSTEM

Automatic robot welding machine is to achieve automatic welding on cable tray production. It will highly improve the welding efficiency, stabilize the welding quality while improving the welding speed. Its composed of three parts;

- Robot System
- Welding System
- Series of Auxiliary Equipment.

ADVANTAGE OF AUTOMATIC ROBOT WELDING MACHINE



- Improve production efficiency, the robot can work continuously without pause or rest in the operation process. Shorten the period of product transformation, It can realize the welding automation of small batch products.
- Improve product quality, with automatic robot welding, the welding parameters of each weld are constant, and the weld quality is less affected by human factors, also reduces the requirements for workers' operation technology.
- Improve the worker's Labor intensity, reduce enterprise cost, one robot can replace 2 to 4 industrial workers, which is different according to the specific situation of the enterprise.
- Easy to arrange production plan, robot welding product cycle is clear, easy to control product output. Quantification the production.
- Kiande is focus on industrial welding robot. Provide customized solutions for free. Fixing before welding, protection during welding, and optimization of corner welding process.

SURFACE ROGHNESS TESTER

A Roughness tester is used to quickly and accurate determine the surface roughness of a material.
Few industries include automobile manufacturing and spare parts processing, machinery parts processing, metal processing equipment & accessories, surface engineering treatment, mold casting, and precision manufacturing.

FEATURES

- Multi-parameter measurement of Ra, Rq, Rz, and Rt.
- Back-lit LCD display with an indication for proper probe alignment.
- Data storage and recall.
- Metric/Imperial conversion.
- Compact and low volume structure for convenient storage.

IMPACT TESTING MACHINE



Impact testers are designed to determine the strength of a sample under impact load. This information is important as this strength determines the performance characteristics of the material under impact loading conditions. Interchangeable strikers, adjustable pendulum drop angle and easy to mount test specimens in the specimen holders, make Impact tester user friendly and measure the impact resistance of a variety of materials in both lzod and Charpy methods. The basic instrument for conducting both the test methods are same. However the attachments and mounting arrangements for these tests are specific to the tests.

MICRO HARDNESS TESTER

Micro Hardness testers provide dependable solutions for precise microhardness testing in quality control and metallurgical research. With testing capabilities ranging from 0.01-2kgf, the VH1102 features a four-position turret for single-scale testing, while the VH1202 offers greater versatility with a six-position turret. Both units support measurements using the digital eyepiece or an optional integrated, high-resolution camera with DiaMet software.

FEATURES

- Precise indenting system paired with a high-quality optical system for unparalleled images
- Optional digital camera housed internally for reliability, protected from dust and misalignment
- Menu-controlled automatic load selection eliminates the need for a load selector knob
- Wide load range (10gf to 2kgf) with 9 individual load steps, streamlining testing without changing internal mechanics.



METALLURGICAL MICROSCOPE WITH REFLECTED, TRANSMITTED LIGHT, CMOS CAMERA



Metallurgical Digital Microscope is designed for studying objects in reflected light using bright field, dark field, and polarization microscopy techniques. This microscope is used to examine metal and semiconductor samples as well as other flat and polished samples with high reflectivity. In transmitted light, you can study translucent samples like thin films and particles trapped on filters. The microscope can be used in metallurgical and mechanical engineering industries, in engineering and quality control laboratories as well as in technical universities.

METALLOGRAPHIC OPTICAL MICROSCOPE

The metallographic microscope system is an organic combination of a traditional optical microscope and computer (digital camera) through photoelectric conversion. It can not only make microscopic observations on the eyepiece but also observe real-time dynamic images on the display screen of the computer (digital camera), and computer-based metallurgical microscope and can edit, save and print the required pictures.

Features including dual incident and transmitted light, high magnification for detailed analysis, polarisation for stress patterns, dark field illumination for contrast, and long working distance objectives. They are used in quality control, research, education, forensic analysis, and industrial applications to examine metal structures and ensure material performance and integrity.



THERMOCYCLERS

Thermocyclers are integral to life science research. In molecular biology they are used for DNA sequencing, cloning, generation of probes, quantification of DNA and RNA, studying patterns of gene expression, detection of sequence-tagged sites, and many more techniques.

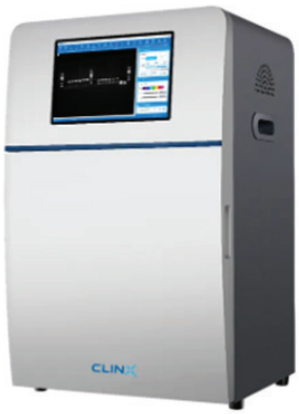
Thermocycler use extends beyond simple PCR and the amplification of nucleic acids to include both random and site-directed mutagenesis as well as the in vitro construction of recombinant DNA sequences. The fine temperature control and ability to hold a precisely set temperature with little fluctuation is also applicable to other techniques in which high-accuracy temperature control increases reproducibility between experiments.

A thermocycler can be used in protocols where very strict temperature control is required such as heating tissue slides for in situ hybridization. The controlled temperature ramping provided by a thermocycler enables the study of temperature-dependent kinetics; for example, in addition to optimizing a PCR protocol, a thermocycler with a gradient feature allows for the determination of the optimum temperature for any enzymatic activity.



GEL DOCUMENTATION SYSTEM

A gel documentation system (also nicknamed gel doc, gel image system or gel imager), is a piece of capital equipment used in molecular biology for the documentation and imaging of nucleic acid and proteins to gather more information on the genetic information stored in DNA.
A gel doc is used to image and document the genetic information from nucleic acids and proteins suspended within polyacrylamide or agarose gels.



BENEFITS OF USING A GEL DOCUMENTATION SYSTEM FOR DETECTING PROTEINS

- Saves time and can reduce overhead cost
- Less exposure to harmful chemicals
- Gives quantitative data
- Supports fluorescent detection for multiplexing
- Produces higher-resolution images
- Ability to access more applications (such as visible gel imaging, Western blotting, total protein normalization)
- Able to image more stains
- Advanced software makes it convenient to save and share data

GAS CHROMATOGRAPHY

Gas chromatography (GC) helps scientists ensure the safety, quality, and purity of our food, air, water, and pharmaceuticals; determine the composition and consistency of fuels and chemicals; and test for controlled substances in criminal investigations and sporting competitions.

Agilent gas chromatography systems combine innovative technology, Instrument Intelligence, and legendary reliability to meet the needs of analytical testing and research laboratories across a broad range of applications, such as environmental, food safety, forensics, energy & chemicals, and more. Integrated Instrument Intelligence helps avoid GC problems before they affect chromatographic performance to optimize uptime, cost of ownership, and confidence in results.



ELISA

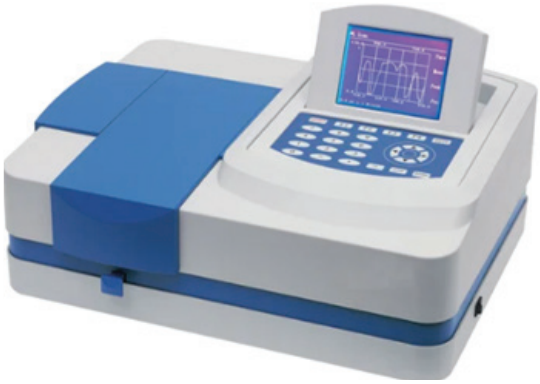
ELISA (enzyme-linked immunosorbent assay) is a plate-based assay technique designed for detecting and quantifying soluble substances such as peptides, proteins, antibodies, and hormones. Other names, such as enzyme immunoassay (EIA), are also used to describe the same technology. In an ELISA, the antigen (target macromolecule) is immobilized on a solid surface (microplate) and then complexes with an antibody that is linked to a reporter enzyme. Detection is accomplished by measuring the activity of the reporter enzyme via incubation with the appropriate substrate to produce a measurable product. The most crucial element of an ELISA is a highly specific antibody-antigen interaction.

ADVANTAGES

- Quick because only one antibody and fewer steps are used.
- Cross-reactivity of secondary antibody is eliminated.

UV VISIBLE SPECTROPHOTOMETER

Ultraviolet-visible spectroscopy or ultraviolet-visible spectrophotometry (UV-Vis or UV/Vis) refers to absorption spectroscopy in the ultraviolet-visible spectral region. This means it uses light in the visible and adjacent (near-UV and near-infrared (NIR)) ranges. UV/Vis spectrophotometer is used in the quantitative determination of concentrations of the absorber in the solutions of transition metal ions and highly conjugated organic compounds.



- Compact and robust
- High-speed measurement at 5 msec.
- Maximum 1,000 consecutive spectrum measurement
- Synchronized measurement function from an external signal

HPLC



HPLC is an analytical chemistry technique applied to separate, identify, and quantify semi- and non-volatile compounds in liquid samples. Industries like pharma and biopharma, environmental, and food and beverage use HPLC for the characterization, purification, and quantitation of compounds.

EVERY HPLC ANALYSIS INCLUDES THE FOLLOWING STEPS

- Mobile phase begins to flow. The pump pushes the eluents or solvents through the system at a specified flow rate.
- Sample injection. Once injected into the mobile phase flow path, the sample travels with the mobile phase from the injection point to the head of the column.
- Compound separation. Physical separation of the compounds happens on the column stationary phase. After elution from the column, the separated sample components travel to the detector.
- Analyte detection. Detection of target analytes based on an electrical signal generated by specific properties.
- Chromatogram generation. Translation of the detected analyte signal by the CDS into a chromatogram of analyte signal versus time.

FTIR

Fourier transform infrared (FTIR) spectroscopy is a powerful analytical technique used to analyze materials across a broad range of industries, including pharmaceutical, polymer/petrochemical, specialty chemical, semiconductor, environmental, forensics, and material science. Adaptable to any workflow, key uses of our FTIR instruments include chemical compound identification, quantitative analysis and QC/QA of chemical blends, analysis of gases for semiconductor fabrication or purity, identification of trace impurities, and more. FTIR microscopes extend analysis capabilities down to minute particles for contaminant analysis or microplastic applications.



DISSOLUTION APPARATUS

Dissolution apparatus is a modular system designed for manual or automated dissolution testing. The instrument can be configured for use with baskets (Apparatus 1), paddles (Apparatus 2), paddle over disk assemblies (Apparatus 5), and rotating cylinders (Apparatus 6), and can accommodate vessel sizes from 100 mL to 2 L.

FEATURES

- Automated or manual dosage delivery options, in-vessel temperature monitoring, and sampling options.
- Interchangeable dissolution accessory components to maintain the correct height.
- Touch panel display enables easy navigation.
- Language options include English, Spanish, Japanese, and Simplified Chinese.
- 708-DS TruAlign vessels ensure correct vessel alignment.
- Easy priming heater/circulator sits behind water bath to save valuable bench space with minimal vibration.
- Decrease cleaning time with the angled water bath—makes cleaning easy and efficient.
- Integrate with Agilent Dissolution Workstation Software for automated control of up to four systems from one PC.
- Online analysis via the Agilent Cary 60 UV-Vis Spectrophotometer and Agilent Cary WinUV Dissolution Software.



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